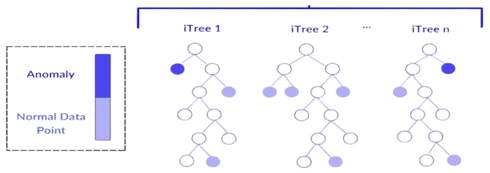
Detecting Outliers in Multiclass Stellar Populations via Isolation Forest Anomaly Detection Frameworks

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INTRODUCTION

- **Stellar anomalies:** Stellar objects that appear or behave differently from what is expected
  - Identifying outliers is a constant struggle due to the complexity and magnitude of vast stellar datasets
- **Morgan Keenan classification system:** System to classify stars based on temperature and spectral characteristics
  - Visualized through **Hertzsprung-Russell** (HR) Diagrams
- **Isolation Forest (IF):**
  - Unsupervised machine learning model
  - Utilize **iTrees** to recursively partition data points and find isolated outliers based on calculated path length
  - Works well with high-dimensional data
- **Stellar Physical Features**: calculated traits of a star



LITERATURE REVIEW

Isolation Forest (IF) has been widely used in anomaly detection due to its efficiency, scalability, and ability to handle high-dimensional datasets without requiring extensive parameter turning. IF is effective when anamolies are few and distinct but may struggle when anamolies are close to the normal class.

- **Liu et. al. (2009)**: utilized a random forest (RF) algorithm for classification of celestial bodies
  - **Results:** achieved a 98% accuracy rate with the low training loss
- **Hariri et. al. (2021)**: Extended Isolation Forest (EIF) utilizes random slope hyperplanes, allowing better isolation of anomalies in datasets with correlated or rotated feature spaces
  - **Results**: EIFs enhanced traditional IFs' accuracy and robustness across both synthetic and real-world datasets, including astronomy-related examples

RESEARCH OBJECTIVES

1. Collect and generate a **processed and classified multi-stellar dataset** with labeled Morgan Keenan classes and calculated physical feature collumns
2. Design an IF model that is able to **correctly identify minority and anomalous stars** within multivariate stellar datasets

METHODOLOGY

Data Acquisition

Data obtained from the **Gaia Data Release 3 catalog**

- 500,000 stars queried

Features:
- parallax
- parallax error
- abs_g_mag
- phot_g_mean_mag
- phot_bp_mean_mag
- bp_rp for each star

Data Preprocessing

- Rows with missing or invalid values were removed
- Stars with parallax errors greater than 20% were filtered

Spectral classes:
- O, B, A, F, G, K, M
- Teff was used to categorize elements into their specific spectral type

Luminosity classes:
- main sequences, giants, white dwarfs, subgiants
- log_L was used to estimate the points' luminosity

Data Classification

Each element was classified based on the **Morgan Keenan** system

- Data preprocessing decreased the dataset from **500,000 elements to 98,500 elements**
- The cleaned and categorized data was then p**lotted on an HR diagram**, with the luminosity classes visualized as varying colors

Model Creation

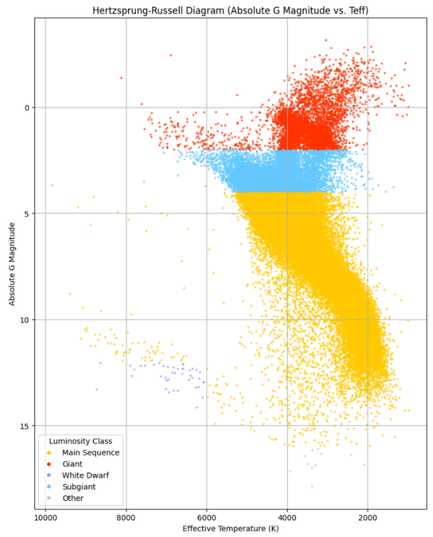
Input
- bp_rp (color)
- Log of luminosity
- Teff (Temperature)

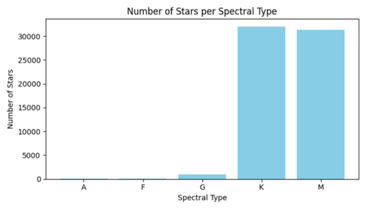
Output
- Anomaly Score

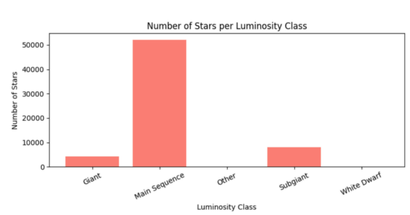
Altered hyperparameters:
N_estimators IF algorithm used 500, silhouette score was calculated with 100,200, and 500
Contamination Used a score of 0.05, calculated silhouette score with values from 0.005 to 0.1

Constant hyperparameters:
Sample size Before cleaning dataset ~500,000 stars, after cleaning dataset ~98,500
Random state Used a score of 42, so results are reproducible and consistent.

RESULTS







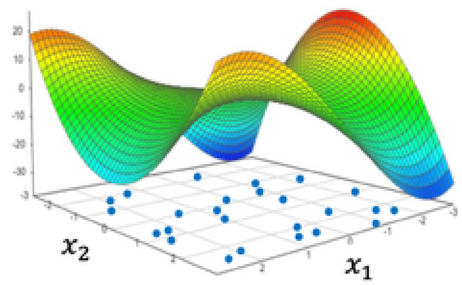

```
Average Silhouette Score for n_estimators=500, contamination=0.005: 0.6191
Average Silhouette Score for n_estimators=500, contamination=0.01: 0.5637
Average Silhouette Score for n_estimators=500, contamination=0.05: 0.3882
Average Silhouette Score for n_estimators=500, contamination=0.1: 0.3290

Top Average Silhouette Scores:
   n_estimators  contamination  average_silhouette_score
0           500          0.005                  0.619136
1           500          0.010                  0.563657
2           500          0.050                  0.388206
3           500          0.100                  0.328974
```

CONCLUSION

- **Relationships & Discovery**:
  - The IF model's accuracy improves by incorporating parallax alongside Teff and log_L, and testing various feature combinations
  - Heteroscedastic errors in Gaia data necessitate Monte Carlo feature sampling to avoid misclassifying faint stars as anomalies
  - This sampling method generates synthetic feature sets using Gaia's uncertainties, enabling more reliable anomaly scoring
  - Anomaly detection reveals stars outside known classifications, suggesting new stellar types
  - Clustering similar anomalies may uncover hidden astrophysical relationships
  - Rare stellar features, such as atypical white dwarfs, can be identified and studied further

- **Future Work**:
  - Explore additional input features and refine sampling techniques
  - Investigate the physical causes behind detected anomalies
  - Apply the model to broader datasets to validate and expand classification boundaries



Monte Carlo:
A random sampling scheme

ACKNOWLEDGMENTS & REFERENCES

Hariri, Sahand, et al. "Extended isolation forest." IEEE Transactions on Knowledge and Data Engineering, vol. 33, no. 4, 1 Apr. 2021, pp. 1479–1489, https://doi.org/10.1109/tkde.2019.2947676.

Liu, Fei Tony, et al. "Isolation Forest." 2008 Eighth IEEE International Conference on Data Mining, Dec. 2008, pp. 413–422, https://doi.org/10.1109/icdm.2008.17.

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